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SCIENCE AND TECHNOLOGY, IGUORIAKHI,
NIGERIA***

***DEPARTMENT OF AGRICULTURAL
TECHNOLOGY***

COURSE CODE: AGT 211

***COURSE TITLE: PASTURE AND FORAGE
PRODUCTION***

COURSE UNIT: 4

Week	Objectives
1 - 3	Understand the general classification, identification and botany of important forage crops.
4 - 5	Understand how to establish pastures and forage crops.
6 - 8	Understand the improvement and management practices of pastures and forage crops
9 - 10	Understand how to make good quality hay and silage.



GENERAL CLASSIFICATION, IDENTIFICATION AND BOTANY OF IMPORTANT FORAGE CROPS

What are Pasture and Forage crops?

Pasture is defined as an area of land covered with grasses, herbaceous legumes, forbs, shrubs, and trees used for livestock feeding or environmental protection. This area is commonly referred to as grassland. It can also be defined as an area of land occupied with forage plant species, whether natural or planted by humans, with the purpose of providing qualitative feed in the right quantity to livestock. The term "pasture" can also refer to the action of feeding, such as the grazing or browsing of animals on pasture plants.

Forage refers to any above ground plant material used for feeding livestock, excluding concentrates and other industrial by-products.

Forage crops are plants cultivated for their vegetative part and are used for the feeding of livestock. Some plants have highly nutritious stems, leaves and shoot which supplies required nutrients for ruminant animals. Examples of such plants include: Elephant grass, giant star grass, carpet grass, centrosema, guinea grass, stylo, etc. The vegetative parts are usually consumed as a whole before they flower. It specifically includes vegetative plant parts, such as the leaves, flowers, and stems of plants that are consumed by domesticated or wild animals.

Pastures and forages constitute the primary sources of feed for different livestock categories worldwide. They are particularly important for ruminant animals, enabling these animals to produce useful outputs from natural resources that humans cannot consume directly.

Pastures can be categorized as:

- **Natural or native pastures** which occur naturally in rangelands, along river banks, or in abandoned lands.
- **Artificially sown pastures** found in commercial farms, research institutes, and colleges of agriculture.

A related term often used is **fodder**, which refers more specifically to the vegetative part of cultivated forages or crops utilized as feed. However, the sources note that it is sometimes difficult to differentiate precisely between feed, forage, and fodder, especially when animals consume non-vegetative parts like the seed head along with the herbaceous plant parts.

Properties of a natural pasture

- They contain poor quality grasses and legumes that may not be suitable for grazing animals
- They soil is usually low in fertility
- Several variety of grasses and legumes are found in it
- Productivity is very low but it is drought resistant
- Most of the species are hardy and cannot be easily eradicated

- It has good regenerative ability and bush burning might be employed in stimulating its re-growth.

Properties of artificial pasture

- They are intentionally sown with high nutritious legumes and grasses
- It contains no weed but some trees that serves as shed
- It has high regenerative ability after grazing
- They can withstand trampling.

Properties of a forage crop

Many plants are edible, but not all of these plant could be referred as a forage crops. A proper forage crop should have the following properties:

- They should supply the nutrients required by the foraging animals
- They must regenerate fast after being browsed
- They should withstand trampling
- Browsing animals should prefer it for its palatability
- It must have a high leaf to stem ratio
- It must have moderate moisture content.

Importance of pasture and forage production

1. Meeting Nutritional and Feed Demands

- Pastures and forages constitute the main sources of feed for different categories of livestock worldwide, making them the backbone of the livestock industry.
- Forages and fodder are essential sources of nutrition for ruminant animals (such as cattle, sheep, and goats) because these animals can produce useful outputs from natural resources that humans cannot consume directly. These animal products, in turn, provide essential nutrients for human existence.
- Proper production and conservation of pastures are necessary to meet the nutritional demands of livestock, including cattle, small stock (sheep), and camels, especially during extreme drought situations.
- Improved pastures, which have undergone breeding and selection, produce higher biomass of superior quality compared to native pastures, enhancing livestock performance. Legumes are particularly valuable due to their relatively high nitrogen content (protein) in the vegetative matter, serving as an important livestock feed resource in drylands.
- Forage conservation techniques, such as making hay or silage, allow farmers to conserve excess forage during periods of abundance for utilization during periods of scarcity, reducing reliance on expensive concentrate feeds.

2. Economic and Livelihood Benefits

- Pasture production contributes to sustainable and profitable livestock production.

- Rearing animals on pastures generally reduces the cost of production for farmers.
- Pasture establishment, utilization, and management help to boost the income level of farmers and improve livelihoods. Farmers can generate extra income through the sales of harvested forage or seeds.
- Products from pasture-finished livestock are higher in beneficial nutrients like omega-3 and conjugated linoleic acids, which are associated with reduced blood cholesterol and cancer risks.
- The sector also provides employment opportunities for people, particularly in forage conservation and marketing.

3. Environmental and Land Management Roles

Pasture and forage management are vital for environmental sustainability and land protection:

- Pasture is defined partly as land used for environmental protection.
- Pastures help to protect and conserve the soil, preventing damages and controlling soil erosion.
- Pastures are essential for site stabilization in areas like dams, under bridges, and in lawns.
- Pasture production and reseeding are critical interventions that contribute to building the resilience of pastoral households, especially in areas affected by drought and climate change.
- The study and management of pastures and forages enable an understanding of the relationship between ecosystem components (soil, plants, animals, man, climate), promoting methods for conserving and protecting the environment from destruction through poor management.
- Pasture species can be integrated with other crops, which helps maintain soil health by providing extra nitrogen (from legumes) and helps combat climate change.
- Pastures established permanently help to retain the structure of soils due to the underground storage organs like rhizomes and stolons.

Classification of forage crops

Forage crops and pasture plants are classified based on several criteria relating to their botanical characteristics, duration of life cycle, usage, and morphological structure.

1. Botanical and Structural Grouping

Forage crops are broadly classified into two basic groups based on their structure:

- **Herbaceous Plants:** These species generally show an annual cycle of growth and decay, although perennial herbaceous species regrow from existing root stock.
 - **Grasses:** These belong to the family Gramineae (monocots). They make up the bulk of plants in natural vegetation that supply animal feed. They are characterized by cylindrical, often hollow stems, leaves with parallel venation, fibrous rooting systems, high energy content, and typically low crude protein. Grasses can be perennial (living more than two years) or annual (lifespan less than one year). They have: fibrous root

systems; Stems could be tillers, stolons, Rhizomes; Leaves have Sheath, Blade, Ligule, Auricle; Inflorescence could be Spike, Raceme or panicle.

Examples of grasses include:

▪ Elephant grass	<i>Pennisetum purpureum</i>
▪ Carpet grass	<i>Axonopus compressus</i>
▪ Spear grass	<i>Imperrata cylindrical</i>
▪ Guinea grass	<i>Panicum maximum</i>
▪ Giant star grass	<i>Cynodon plectostachyum</i>
▪ Northern gamba	<i>Andropogon gayanus</i>
▪ Southern gamba	<i>Andropogon tectorum</i>

- **Legumes:** These belong to three families (Mimosoideae, Caesalpinoideae, and Papilionaceae). They have broad leaves with net venation, and produce seeds in pods. They are essential because they fix nitrogen, leading to a relatively high nitrogen content (protein) in their vegetative matter. They have: root systems (Taproot, Nodule formation, Symbiotic), Stems and Leaves (Pinnate, Palmate, Trifoliate leaves), Flower structure and fruit/seed development (Pods).

Examples of legumes include:

• Centro	<i>Centrosema pubescens</i>
• Stylo	<i>Stylosanthes gracilis</i>
• Calopo	<i>Calopogonium mucunoides</i>
• Muccuna	<i>Muccuna utilis</i>
• Sun helmp	<i>Crotalaria juncea</i>

- **Woody Plants:** These plants grow cumulatively over many years. They include Fodder trees and shrubs.

Examples of woody plants include:

- *Leucaena leucocephala*
- *Gliricidia sepium*
- *Gmelina aborea*
- *Ziziphus mauritiana*
- *Ficus thonningii*
- *Acacia albida*

2. Life Cycle or Duration

Pasture plants are categorized based on the length of time they live and remain productive:

- **Annuals:** Plants that complete their life cycle within one growing season and survive the dry season as seeds.
- **Biennials:** Plants that require two seasons to complete their life cycle; the first season for vegetative growth and the second for reproductive growth.

- **Perennials:** Plants that grow throughout the year for many years, continuously producing flowers and seeds without dying.

Related to life cycle, pastures themselves are classified by duration of use:

- **Temporary Pastures (Leys):** Established as part of a rotation system with the intention of being converted back into farmland, lasting typically three to five years.
- **Semi-Permanent Pastures:** Established due to a break in the cropping phase or as part of a rotation, generally lasting five to ten years.
- **Permanent Pastures:** Established for long periods, lasting several decades, and intended solely for feeding livestock.

3. Usage, Origin, and Quality

Pastures can be classified based on how they are utilized and their origin:

- **Native or Natural Grasslands (Natural Pastures):** The herbage species are not sown or planted. They consist mainly of indigenous species and have evolved in harmony with prevailing soil and climatic factors. These are generally low in quality and have short growing seasons compared to improved species.
- **Improved Pastures:** Pastures that have undergone breeding, selection, and improvement for better yield and livestock performance. They are often exotic or introduced species that produce a higher biomass of superior quality.
- **Pastures Associated with Crops:** Pasture species incorporated into other crops, such as field crops (e.g., cereals) or tree crops (e.g., coconut plantation).

4. Consumption Habit

Pasture plants can also be distinguished by how livestock consume them:

- **Grazed Plants:** Short herbs consumed with both the leaves and stems together.
- **Browed Plants:** Woody plants where only the leaves and soft stems/twigs are eaten, leaving the woody stem uneaten.

Factors Affecting Nutritional Value and Quality of forage crops

The nutritional quality of pasture largely determines its value as feed for livestock, particularly the content of crude protein (CP) and digestibility.

1. **Stage of Maturity/Harvest Timing:** This is the most critical factor affecting forage quality.
 - **Decline in Quality:** An extended period between cuttings or advancing plant maturity increases the percentage of dry matter (DM), crude fiber, lignin, and cell wall structure. This increase in fiber reduces intercellular space and causes the midribs and leaf sheaths to attain a greater percentage of fiber and lignin.

- **Nutrient Reduction:** As plants mature, there is a subsequent decrease in the leaf-to-stem ratio, percentages of crucial nutrients like crude protein (CP), mineral constituents (P, K, Ca, Mg), and soluble carbohydrates.
 - **Digestibility and Intake:** More mature herbage is less nutritive and less appealing to grazing animals. Digestibility declines rapidly as the cell wall structure increases, which slows down the passage of feed through the animal's digestive system.
 - **Optimizing Quality:** If the objective is a high-quality forage feed with high CP, harvest should typically be done early, before the seeding stage, although this is at the expense of total DM yield.
2. **Species and Botanical Composition:**
- **Grasses vs. Legumes:** The type of plant profoundly impacts quality. Legumes are essential livestock feed resources in drylands, providing a cheap source of protein due to their relatively high nitrogen content in the vegetative matter. Grasses, conversely, are typically higher in energy but lower in crude protein.
 - **Palatability:** Palatability and digestibility are inherent attributes that determine quality and intake. Land degradation can lead to a decline in palatable species and an increase in unpalatable ones.
3. **Fertilizer Application:**
- Nitrogen (N) fertilizer application typically increases crude protein (CP) content in grasses but may depress the concentration of other minerals such as Phosphorus (P), Potassium (K), Calcium (Ca), and Magnesium (Mg). This depression is primarily due to the dilution effect caused by the significantly increased herbage production (yield).
4. **Weeds, Pests, and Diseases:**
- Weeds reduce the overall quality of pasture, especially when they occupy large areas. While some weeds may be nutritious, others can be toxic or poisonous to livestock, posing environmental hazards.
 - Pests and diseases also have adverse effects on pasture quality.
5. **Forage Conservation Methods:**
- The quality of preserved forage (such as hay or silage) depends on the harvest stage, post-harvest handling, storage conditions, and prevailing weather conditions.
 - **Silage** generally maintains quality better than hay. If hay is stored with high moisture content (over 12%), it favors microbial growth (bacteria, yeast, fungi), degrading soluble sugars and causing a decline in quality.
 - **Standing hay** experiences continuous quality deterioration.

Factors Affecting Productivity and Yield of forage crops

Productivity refers to the quantity of herbage or biomass produced.

1. **Climatic and Environmental Factors:**
- **Water/Moisture:** Availability of moisture is a key determinant factor. In arid and semi-arid areas (ASALs), pasture productivity naturally declines both in quantity and quality during dry periods and is optimal during the wet seasons. Drought intensity and frequency, coupled with climate change, severely affect forage supply.
 - **Soil Conditions:** Soil type, fertility, and topography influence pasture production. Poor nutritional status of the soil restricts the animal carrying capacity, thus lowering

effective output. Pastures established on fertile soils that are relatively flat typically yield better.

2. **Management and Agronomic Practices:**

- **Fertilizer Use:** The application of fertilizers can increase the output of grasslands. **Nitrogen (N)** fertilization dramatically increases the dry matter (DM) production of grasses. The goal of fertilization is to supply limiting nutrients identified through soil analysis.
- **Species Selection:** Choosing species adapted to the prevailing ecoclimatic zones (rainfall, temperature, radiation) and soil types is crucial, as this determines the end productivity of biomass and seeds. Using **multiple species** is generally preferred over monocultures, as diversity helps increase resilience to environmental shocks and may increase yield stability.
- **Establishment Practices:** Proper timing for reseeding (to capitalize on the first rains) and the choice of sites with deep soils to support soil water moisture significantly influence biomass yields. Land preparation is critical as it loosens the soil, enables better infiltration of rainwater, and removes competing weeds, giving the target crop a head start.

3. **Utilization and Grazing Management:**

- **Defoliation:** The frequency and intensity of defoliation greatly affect herbage production and regrowth potential. Severe and prolonged intensive grazing can injure most plants, which reduces organic synthesis and adversely affects the regrowth capacity of the pasture.
- **Stocking Rate:** This is the most important factor affecting forage utilization, as it establishes a strong interaction between forage availability and defoliation/intake. Overgrazing, deforestation, and poor grazing practices contribute to land degradation and loss of primary productivity.

4. **Competition and Hazards:**

- **Weed/Bush Encroachment:** Weeds cause competition for nutrients, light, space, and water, reducing productivity. Bush and weeds encroachment can worsen the availability of forage.
- **Pests and Diseases:** Pests and diseases have an effect on dry matter productivity and yield.

ESTABLISHMENT OF PASTURES AND FORAGE CROPS

The establishment of pasture and forage crops is a critical process, encompassing a series of activities from site selection and land preparation to the actual sowing or planting of the chosen species. Successful establishment is necessary for maximizing biomass and seed yields and for the long-term survival of the pasture, especially in challenging environments like the arid and semi-arid areas (ASALs).

Importance of establishing pastures and forages

Pasture and forages are established for a combination of reasons related to **livestock production, economic benefit, and environmental protection and remediation.**

1. Sustaining Livestock Production and Meeting Nutritional Demands

The core reason for establishing pastures and forages is to support the livestock industry, particularly in areas like the arid and semi-arid areas (ASALs).

- **Primary Feed Source:** Pasture and forage crops are the primary sources of feed for different categories of livestock worldwide, and the backbone of the Nigerian livestock industry.
- **Nutrient Supply:** Forages provide essential nutrition for ruminant animals (like cattle, sheep, and goats), allowing them to produce useful outputs from natural resources that humans cannot consume directly, thus supplying essential nutrients for human existence.
- **Drought Resilience:** Establishing pasture and reseeding interventions are vital for building the resilience of pastoral households. This is necessary to meet the nutritional demands of cattle, small stock (sheep), and camels, especially in extreme drought situations.
- **Superior Quality:** Improved pastures, which have undergone breeding and selection, produce higher biomass of superior quality compared to native pastures. Legumes, in particular, provide a cheap source of protein due to their high nitrogen content, which is valuable for livestock feed.

2. Environmental Protection and Restoration

Pasture establishment is crucial for protecting land health, especially in degraded areas:

- **Combating Land Degradation:** Pasture production and reseeding is a critical intervention for restoring degraded sites, often involving the reseeding with native species. Land preparation for pasture establishment also helps to remove unwanted plants (weeds).
- **Soil and Site Conservation:** Pastures are established to protect and conserve the soil, preventing damages and controlling soil erosion. They are also used for site stabilization in areas like dams, under bridges, and in lawns. Land preparation activities, such as incorporating micro-catchments and terracing, aim to reduce runoff and enable better infiltration of rainwater.
- **Mitigating Climate Impacts:** Establishment helps communities cope with the increasing impacts of droughts coupled with climate change, which have adversely affected forage supply.
- **Maintaining Soil Health:** Integrating pasture species with other crops (undersowing) helps maintain soil health, especially when legumes are included in the rotation system to provide extra nitrogen and improve soil fertility.

3. Economic and Management Benefits

Establishing pastures supports the long-term profitability and functionality of the farm:

- **Profitability and Cost Reduction:** Establishing and managing pastures contributes to sustainable and profitable livestock production. Rearing animals on pastures helps reduce the cost of production.
- **Income Generation:** Pasture production helps to boost the income level of farmers and improve livelihoods. Farmers can generate extra income through the sales of harvested forage or seeds.
- **Soil Fertility Restoration:** Temporary pastures, known as leys, are frequently included in crop rotation specifically for the purpose of restoring soil fertility. Legumes or grass/legume mixtures are generally preferred for this role.
- **Pest and Disease Control:** Temporary pastures can be established to help farmers control weeds, disease pathogens and pests as part of a rotation system.

Methods establishing pastures and forage crops

The establishment of pasture and forage crops involves sequential operations designed to ensure the successful germination, survival, and maximum production of the desired plant species. These steps span pre-planting planning, physical land preparation, sowing, and immediate post-planting management.

1. Pre-Planting Operations: Selection and Timing

The process begins with critical decisions regarding the location and timing of the establishment.

- Site Selection:** The chosen site should support quick establishment and have good soil characteristics. Key considerations include selecting areas with deep soils to support soil water moisture and areas that are fairly flat to reduce the possibility of seeds being washed away by runoff.
- Timing:** Proper timing is crucial, particularly in rainfed systems like arid environments, as planting must be done early to capitalize on the first rains. In some areas, the short rains (October to December) are optimal for rangeland grass species.
- Species Selection:** Species must be chosen based on their adaptation to prevailing ecoclimatic zones (e.g., rainfall, temperature) and soil types. Preference should be given to native plants over non-native species, as non-natives often become invasive weeds. It is generally preferred to use multiple species or species with multiple functional traits rather than monocultures to increase diversity and resilience.
- Seed Treatment:** Before sowing, seeds may need treatment, such as applying fungicides and insecticides, or using hot water (scarification) to break seed dormancy. The appropriate seed rate and sowing depth must be determined, typically shallow for grasses (2 cm) and medium for legumes (5–7 cm).

2. Land Preparation

Land preparation is considered **very critical for the success** of pasture or sward establishment.,,

- i. **Land Clearing:** The land must be cleared, which can be done manually, chemically, mechanically, or through burning. This process, especially bush clearing, should be done carefully to prevent exposing the land to erosion.
- ii. **Seedbed Conditioning:** A primary objective is to create a seedbed that is weed-free, fine, firm, moist, leveled, and well-drained. This typically involves steps such as ploughing, harrowing, leveling, and constructing ridges for large-seeded legumes. Fine and firm seedbeds are necessary to ensure small seeds make good contact with the moist soil particles,.
- iii. **Water Harvesting and Soil Conservation:** Land preparation must incorporate methods to increase water harvesting to benefit the crops. Good land preparation loosens the soil, enables better infiltration of rainwater, reduces runoff, and helps control soil erosion. Techniques for water harvesting and conservation include contour bunds, Fanya juu terracing, Zai pits, Tumbukiza, and Jembe scratching.

3. Sowing and Planting Methods

Pasture seeds are often small, necessitating careful placement and covering to enhance germination. Methods vary based on the scale of production and the type of material used:

- i. **Broadcasting:** Seeds are scattered manually or by machine. Due to the small size of the seeds, they may be mixed with inert material like sand or sawdust to ensure uniform spreading,. The seeds must be lightly covered afterward, for instance, by raking or pulling tree branches over the area,.
- ii. **Seed Drilling:** This is recommended for large-scale reseeding as it covers the seeds immediately after placement and allows the operator to set the seeding depth,.
- iii. **Oversowing or Sod Seeding:** This involves improving an existing pasture by seeding over it with improved forage material. The existing pasture should be lowered (e.g., through grazing or burning) to reduce resource competition for the new seeds. This method avoids disturbing the soil, thereby reducing erosion.
- iv. **Vegetative Propagation:** For grasses that produce few or unviable seeds (e.g., Napier grass or Star grass), establishment is done using stem or root cuttings or splits, ensuring that 2–3 nodes are buried in the soil.
- v. **Undersowing:** Pasture species are integrated with other crops (such as cereals) either as an intercrop or cover crop to maximize productivity per unit of land.
- vi. **Furrow Line Sowing:** Seeds are sown lightly along contour furrow lines and covered with a thin layer of soil, which helps to prevent erosion,.

4. Post-Planting Operations

Once the pasture germinates, management practices are necessary to ensure successful establishment and optimal yield:

- i. **Weeding:** Timely weeding is essential to give the desired crop a head start by removing unwanted plants that compete for nutrients, light, space, and water,., Weeds can be controlled manually (hand hoes), chemically (herbicides), or mechanically,.

- ii. **Fertilizer Application:** Fertilizers may be applied after weeding to improve performance. Nitrogenous fertilizers, often limiting in tropical soils, may be applied in split doses for grasses, while phosphatic fertilizers can be applied during land preparation,.
- iii. **Defoliation:** The pasture can be defoliated (cut) to stimulate tillering, increase productivity, and improve the growth performance and quality of the forage,. This may be done 2–3 times a year depending on the species and environmental conditions.

Artificial/Sown Pasture

Pasture and forage crops can be established as **pure stands** (sole species) or **mixed pastures** (such as grass/legume associations), and each system has distinct advantages and disadvantages related to nutrition, stability, and management complexity,.

1. Pure Stands (Sole Species)

Pure stands consist of planting a single species, such as a sole grass or a sole legume.

Disadvantages of Pure Stands (Grass Monocultures):

- i. **High Fertilizer Dependency:** Sole grass pastures often require heavy applications of nitrogen (N) fertilizer to produce high yields and adequate protein content,.
- ii. **Cost and Resource Use:** Supplemental plant nutrients are typically required for optimal output, which necessitates the use of external resources like fertilizer.
- iii. **Restricted Diet:** In intensively managed farms, livestock may be restricted to feeding on only one type of pasture species without alternative choices.

2. Mixed Pastures (Grass/Legume Mixtures)

Mixed pastures, typically combining grasses and legumes, are fully sown pastures that belong to the artificial grasslands or leys category.

Advantages of Mixed Pastures:

- i. **Nutritional Improvement:** Legumes have a high value for animal production primarily due to their high nitrogen content (protein) in the vegetative matter, which is used as animal feed.
- ii. **Soil Fertility and Sustainability:** Legume components possess the ability to fix nitrogen. Grass/legume mixtures are generally preferred for restoring soil fertility when incorporated into crop rotation systems.
- iii. **Increased Resilience and Yield:** Using multiple species or species with multiple functional traits is often preferred over monocultures because it increases the pasture's resilience to environmental shocks and enhances the diversity of diets available to livestock.
- iv. **Ecological Benefits:** Mixed pastures provide habitat for biodiversity.

Disadvantages and Management Challenges of Mixed Pastures:

- i. **Management Complexity:** The management of mixtures is difficult because grasses and legumes have different vegetative and reproductive cycles.
- ii. **Competition:** The botanical composition of the mixture is strongly influenced by competition for nutrients, water, and light. Tropical grasses are often more aggressive and extract more nitrogen (N) from the soil than legumes.
- iii. **Legume Decline due to Fertilization:** The continued use of high N fertilizer, which grasses require for high yield, causes a rapid decline in the legume component of the mixture. Applied nitrogen reduces the nodulation of the legume, further interfering with its competitive capability within the sward.

IMPROVEMENT AND MANAGEMENT PRACTICES OF PASTURES AND FORAGE CROPS

Pastures- These are grazing lands dominated by grasses and legumes.

Forage crops- These are cultivated plants harvested for fresh feeding, hay or silage.

Improved pastures, by contrast, use selected high-yielding varieties. In April 2025, after 48 years, the National Committee on Crop Varieties released eight new varieties through the National Animal Production Research Institute (NAPRI). These include

Grasses

Brachiaria ruziziensis- Bracharia

Chloris gayana- Rhodes grass

Sorghum almum- Sudan grass

Andropogon gayanus- Andropogon

Panicum maximum- Guinea grass

Pennisetum purpureum (Napier grass)

legumes –

Mucuna pruriens- velvet bean

***Lablab purpureus*- white Lablab/Shika Legume 2**

These varieties offer higher biomass (15–30 t/ha/year), better crude protein (8–20 %) and drought tolerance. Grass-legume mixtures are ideal because legumes fix 50–200 kg N/ha/year, improving soil fertility and animal performance.

Establishment follows six clear steps:

1. Site selection – The soil should be well-drained, fertile and with ≥ 700 mm rainfall.
2. Land preparation – Bushes should be cleared, plough and harrowed. Soil test (pH and nutrients) should be done.
3. Species selection and seed treatment – scarify hard-seeded legumes, inoculate with *Rhizobium*.
4. Planting – broadcast method or drill at recommended rates (5–10 kg/ha for grasses, 2–4 kg/ha for legumes), with vegetative propagation for should be done for Napier.
5. Early care – application of basal Phosphorus and potassium should be done. Weed control is done manually or with selective herbicides.
6. First defoliation (treaming) – after 8–12 weeks at 10–15 cm height. This is beneficial to the vegetation and helps rapid regrowth and proper spread at maturity

Improvement practices

These include

1. over-sowing legumes into native swards
2. controlled burning in the early dry season to remove dead material and stimulate regrowth
3. fertiliser application (split N for grasses, P for legumes)

Management revolves around;

1. Sustainable grazing.
2. Continuous grazing is simple but causes overgrazing and soil compaction.
3. Rotational (paddock) grazing which is superior and involves dividing land into 4–8 paddocks, with grazing occurring 3–7 days and the land is allowed to rest for 21–42 days. This raises carrying capacity from 1–2 to 4–6 total land unit (TLU/ha) and maintains sward quality. Other systems include zero-grazing (cut-and-carry) for high-value forages and strip grazing.
4. Monitor stocking rate (match animals to available forage), sward height and botanical composition.
5. Additional practices involves integration with agroforestry (*Leucaena*, *Gliricidia*), ley farming (pasture-crop rotation) and fencing.

Major challenges in establishment cost

1. Erratic rainfall, seed scarcity (now improving via government programmes)
2. Overgrazing
3. Bush encroachment and land-tenure insecurity.

4. Government initiatives such as the 2025 National Pasture Planting Day and grazing-reserve development aim to address these. When properly managed, improved pastures restore biodiversity, prevent erosion and support year-round feeding.

MAKING GOOD-QUALITY HAY AND SILAGE

Forage conservation bridges the 6–8 month dry-season gap. Hay is dehydrated forage (15–20 % moisture) while silage is preserved by anaerobic lactic-acid fermentation (60–70 % moisture, pH 3.8–4.5).

Hay production (best in short dry spells):

1. Select forage – early-bloom stage (*Panicum maximum*, Rhodes grass or Napier) for maximum nutrients.
2. Cut early morning with sickle or mower.
3. Field-dry rapidly – spread in thin swaths, ted/turn twice daily; crush stems to equalise drying.
4. Test readiness – stems snap, leaves crinkle, no green colour, moisture 15–20 % (grab test).
5. Bale or stack under roofed, ventilated barns to prevent mould and termites.

Quality hay indicators

1. Green colour
2. Leafy
3. Pleasant smell
4. High palatability.

Challenges

Tropical challenges include sudden rain and high humidity causing mould and nutrient loss (carotene, protein). Artificial drying or rack-drying helps in humid zones.

Key Steps in Silage Making

1. **Select the Crop and Harvest at the Optimal Stage** Choose a high-quality forage crop with sufficient soluble carbohydrates (sugars) for fermentation. Ideal crops: maize/sorghum (at milk to dough stage), tropical grasses (pre-flowering/early heading), or legumes (early podding).
 - Harvest when the plant is young and nutritious — avoid over-mature material (stemmy, low sugar).
 - In tropics, target whole-plant dry matter (DM) of 30–40% (60–70% moisture) for best compaction and fermentation.
 - Cut in the morning on a dry day to minimize field losses.
2. **Wilt the Forage (If Necessary)** If the crop is too wet (>70% moisture, common in fresh tropical grasses), allow partial wilting in the field for 1–2 days (spread in thin swaths and turn occasionally).

- Goal: Reduce moisture to 60–70% (squeeze test: chopped material forms a ball that holds shape then slowly expands with little/no free juice).
- Wilting improves packing density and reduces effluent (seepage) losses. Skip for maize/sorghum if already at ideal DM.
- 3. **Chop the Forage** Chop the material into small pieces (2–5 cm long, ideally 1–2 cm for maize kernels to crack).
 - Use a forage harvester/chopper for uniform size.
 - Small pieces: improve compaction, exclude air better, increase surface area for bacteria, and enhance fermentation speed.
 - In smallholder systems (e.g., manual or tractor-mounted chopper), ensure kernels/stems are broken.
- 4. **Add Additives (Optional but Recommended in Tropics)** Tropical forages (especially grasses) often have low sugars → add fermentable carbohydrates.
 - Common: Molasses (2–5% by weight), inoculants (lactic acid bacteria), or urea (for nitrogen boost in low-protein forages).
 - Mix evenly during filling to promote desirable (lactic) fermentation and prevent undesirable (butyric/clostridial) spoilage.
- 5. **Fill and Compact the Silo Rapidly** Transport chopped forage quickly to the silo (bunker, trench, pit, plastic bag, or wrapped bale) to minimize air exposure.
 - Fill in thin layers (15–30 cm) and compact immediately and heavily (tractor wheels, roller, or manual trampling).
 - Aim for high density ($>600\text{--}700\text{ kg/m}^3$ fresh weight) to exclude oxygen.
 - Fill continuously without long delays to prevent heating.
- 6. **Seal the Silo Airtight** Cover immediately with plastic sheeting (black/white or oxygen-barrier film), overlap edges, and weigh down with soil, tires, sandbags, or gravel.
 - Seal edges and sides completely to create anaerobic conditions.
 - For bags/bales: Use multiple layers of stretch film.
 - Poor sealing = air entry → mould, heating, and spoilage.
- 7. **Allow Fermentation (Storage Phase)** Let the silage ferment undisturbed for at least 3–6 weeks (ideally 4–8 weeks for full stabilization).
 - Four main phases occur:
 - **Aerobic phase** (first 1–2 days): Residual oxygen used up; slight heating.
 - **Fermentation phase** (days 2–30+): Lactic acid bacteria convert sugars → lactic acid; pH drops rapidly to 3.8–4.5.
 - **Stable phase**: pH stable; silage preserved if anaerobic.
 - **Feedout phase**: Open face daily and remove $\geq 15\text{ cm/day}$ to avoid reheating/spoilage.
- 8. **Feed Out Properly** When ready, open the silo and feed out from a smooth, vertical face.
 - Remove at least 15–30 cm per day (more in hot weather) to minimize aerobic spoilage.
 - Cover the open face nightly with plastic to reduce air exposure.
 - Discard any mouldy/top-spoiled layer.

Advantages of silage over hay in Nigeria: less weather-dependent, higher nutrient retention, utilises wet-season surplus. Challenges: need for chopping equipment, effluent management and risk of clostridial (butyric) fermentation if poorly packed. Prevention: rapid filling, good compaction and additives.

Conclusion and Summary Sustainable pasture improvement and forage conservation are not optional; they are essential for Nigeria’s livestock roadmap, climate resilience and food security. Adopt mixed pastures with 2025-released varieties, practise rotational grazing and conserve surplus as hay or silage. These practices raise livestock productivity, restore land and reduce conflicts.

Differences between hay and silage

No.	Aspect	Hay	Silage
1	Moisture Content	Low (15–20% moisture)	High (60–70% moisture / 30–40% dry matter)
2	Preservation Method	drying	Anaerobic fermentation
3	Weather Dependence	sunny weather for field drying	can be made during wet/rainy periods
4	Nutrient Retention	Moderate losses	Generally better retention of nutrients
5	Suitability	Challenging due to high humidity	More reliable and preferred

Attached Assessment Questions FOR PRACTICE

3 Theory Questions (10 marks each)

1. Outline the six steps of improved pasture establishment in Nigeria and explain why grass-legume mixtures are preferred. (10mks)
2. List 3 grasses and 3 legumes with their scientific names (10 mks)
3. Fill in the missing text in the table below (10mks)

No.	Aspect	Hay	Silage
1	Moisture Content		
2	Preservation Method		
3	Weather Dependence		
4	Nutrient Retention		
5	Suitability		

10 Objective Questions

1. Target moisture content for good hay is: a) 60–70 % b) 15–20 % c) 40 %
2. Silage is preserved by: a) Complete drying b) Anaerobic lactic-acid fermentation
3. Best cutting stage for hay is: a) Mature dry b) Early bloom
4. Ideal silage pH is: a) 6.0–7.0 b) 3.8–4.5
5. In humid Nigeria, silage is preferred over hay because it is: a) More weather-dependent b) Less affected by rain
6. Common additive for tropical grass silage: a) Molasses or inoculants b) Sand
7. Poor silage is identified by: a) Pleasant acidic smell b) Rancid/butyric smell
8. Hay must be stored: a) In open fields b) Under roof/cover
9. Recommended silo face feed-out rate: a) 1 cm/day b) ≥ 15 cm/day
10. Wilted material for silage targets: a) 15 % moisture b) 60–70 % moisture